

IT314 - Software Engineering

Lab Assignment-3 – Automated Pipeline for identifying FR and NFR.

You need to identify the stakeholders, elicitation techniques, identification of functional and non-functional requirements.

1. Background

Requirements Elicitation is one of the most human-dependent stages of the software engineering lifecycle it involves identifying stakeholders, understanding their goals, choosing the right technique to draw out information from them, and finally converting that information into Functional and Non-Functional Requirements.

In this assignment, you will build an **LLM-powered pipeline** that automates this elicitation process for a given system, using LangChain and prompting techniques covered in the previous lab.

2. Task

You are required to build a LangChain-based pipeline that takes the case study as input and progresses it through the following stages of requirements elicitation:

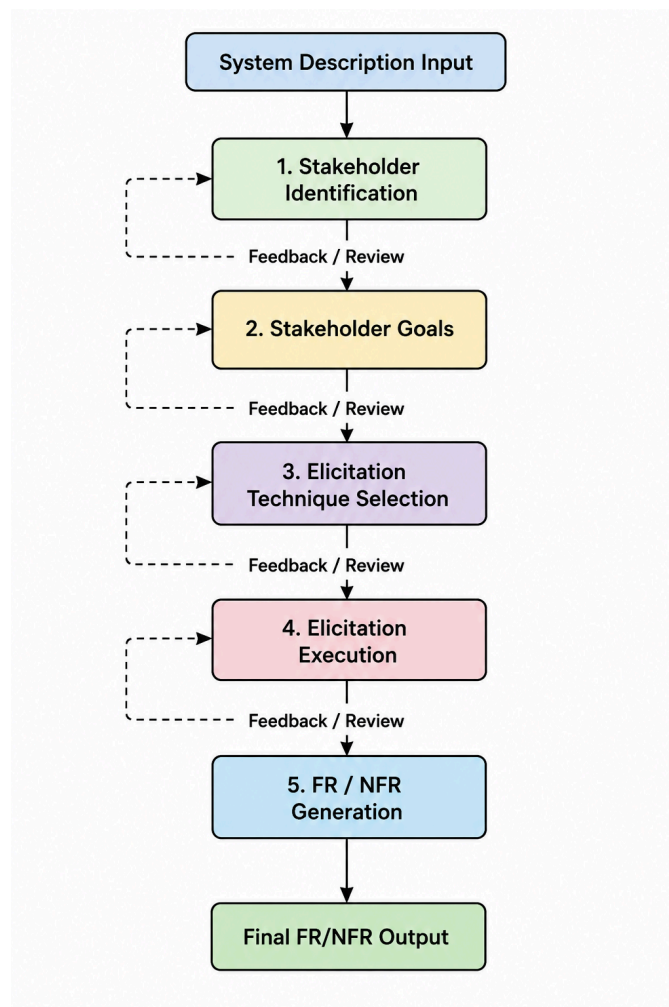
1. **Stakeholder Identification** Identify the relevant stakeholders for the given system.
2. **Stakeholder Goals** for each of the stakeholders carried forward from Stage 1, identify their goals/needs from the system.
3. **Elicitation Technique Selection** Based on the stakeholders and their goals, decide which elicitation technique(s) are appropriate (e.g., interview, questionnaire, workshop, brainstorming, observation, etc.).
4. **Elicitation Execution** Carry out the chosen technique(s) e.g., if a questionnaire is selected, generate the actual questions; if an interview is selected, decide whom to interview and draft the interview questions.
5. **FR/NFR Generation** Using everything gathered in the previous stages, generate a structured list of Functional Requirements (FRs) and Non-Functional Requirements (NFRs) for the system.

Mandatory Design Requirement Feedback at Every Stage

The pipeline **must not** run the above five stages blindly. At each stage transition, there must be **some feedback/review mechanism** that lets a review, edit, approve, or steer the output of one stage before it is carried forward into the next.

What form this feedback mechanism takes is intentionally left for you to design and justify. There is no single correct implementation think about what is practical to build and appropriate for each stage, and be ready to explain your reasoning. Hint: Human-in-the-loop (HITL), ClarifyLLM, LLM as a Judge etc., you can develop for feedback mechanism.

3. Expected Pipeline (Block Diagram)



4. Target System

LIC Market-Driven System

LIC, an insurance company, wants to digitize a range of business processes and provide a complete solution that addresses all aspects of the agent-insurer relationship. Consider yourself as a part of the Requirement Analyst team at Retinodes Software Company, and your job is to gather, analyze, negotiate, validate, and prioritize the set of requirements.

In this new requirement of the project, there are no existing systems that can be analyzed for the development. Requirements have to be gathered, negotiated, validated, and prioritized through multiple stakeholders. This is a complex process because different stakeholders have different perspectives, requirements, expectations, and priorities. Therefore, Retinodes wants to have a requirements engineering framework that can be used in market-facing projects where requirements need to be gathered from multiple stakeholders.

To start with, you need to identify the set of stakeholders associated with the system, understand the domain information about the insurance market, and identify possible system features. Stakeholders may include customers, insurance agents, LIC management,

policy administrators, underwriters, and other relevant personnel. The insurance domain should also be studied to understand existing policies, customer demands, pricing, business rules, and restrictions.

The first product LIC wants you to develop is a system for creating consolidated insurance packages that can compete with packages provided by other insurance companies. Existing insurance packages should be analyzed to understand their features, benefits, pricing, and conditions. The requirements of customers and insurance agents should also be considered while designing these packages.

Another product is based on customer priority. Based on the insurance policies available, the customer can create his/her own package by selecting suitable policies and send a request for review. The system has to automatically analyze the proposed package, identify possible issues or restrictions, provide suggestions (if any), and finally give a competing price for the package.

To understand the problem domain, existing insurance packages have to be analyzed and the demands and restrictions from insurance policies and agents have to be understood completely. Different stakeholders may have conflicting requirements, so the requirements need to be negotiated, validated, and prioritized based on business importance and feasibility.

The requirements and feasibility report generated by you will be further used by the development team for implementation. The report should provide a clear understanding of the stakeholders, domain information, system features, functional and non-functional requirements, constraints, priorities, and feasibility of the proposed solution.

5. Deliverables

- **Working code** implementing the pipeline described above, built using LangChain and LLM API calls.
- **Prompts developed at each stage of the pipeline**
- **Output Screenshots / Output.txt / Json files (Can Submit output in any Form)**
- **A short report** (2–3 pages) explaining:
 - The feedback/review mechanism you designed at each stage, and why.
 - Any notable prompting or design decisions made at each stage.